Falling Dominoes

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1 Abstract

The purpose of this report is to develop a working model that can calculate the topple time of a chain of dominoes accurately. First we applied our theory to a chain of dominoes in the most basic way, calculating the total topple time for a chain of N dominoes spaced evenly. This is considered the most basic set up since it has the least amount of variables and ambiguity. After we determined that our model was able to make predictions accurately, we applied it to several other situations. This forced our model to evolve to a point where it can calculate the topple time of a chain of dominoes in various arrangements. A set of definitions and assumptions used for our theory is also included. The theoretical model's predictions are compared with the experimental results and discussed.

2 Introduction

In this report, an explanation of the phenomena behind a chain of toppling dominoes is given. The understanding developed while studying these dominoes can potentially be applied to natural instances such as the toppling of the Larson B ice shelf, the pattern toppling trees generate in a forest, and disturbances in economic systems. We investigated different scenarios and combinations of toppling dominoes. The theory we developed in order to analyze these chains was altered and extended in order to complete different milestones. This theory was then applied to physical demonstrations in which we competed with other teams comprised of our peers. Our theory has been applied to our model in order to accomplish four goals. First, we wanted to accurately predict the total topple time for a chain of evenly spaced dominoes. Next, we determined how to minimize the topple time for a chain of dominoes that covered a specific distance. Then, we developed a setup for a chain of dominoes whose topple time we predicted with the least amount of error and challenged other teams to predict the topple time more accurately. Finally, we adjusted our model to compensate for non-linear setups. While developing our model, we performed several experiments using a high speed camera in order to gain a control value for the time to compare with the values produced from our theory. The theory provided in this report is based primarily on an article written by D. E. Shaw, [1].

3 Assumptions

The following list includes preliminary assumptions that are pertinent to the understanding of this project. Undoubtedly, there are other assumptions not listed that will be determined as the modeling process progresses.

1. Assume that the air contributes little dampening and can be ignored.
2. Assume that the domino is perfect in shape and of homogenous composition.
3. Assume that all dominoes are identical.
4. Assume that there is no traction between the touching dominoes as they fall.
5. Make this into a 2D problem by assuming the dominoes are of infinite length. This also means that the dominoes are perfectly aligned.
6. Define the force that the domino is exerting on the next domino as a function of θ.

There are several conditions under which we will be performing the experiments and on which we will be basing our theoretical model. These conditions include:

1. The dominoes are parallel to each other and equally spaced.
2. There is no sliding of the dominoes on the plane and each domino remains in contact with the plane at least along one edge.
3. Each domino rotates in one direction only.
4. There are no external friction losses.
5. The dominoes are perfectly rigid with no energy lost at contact.
6. All the dominoes are identical with no variation of its defining dimensions.

4 Definitions

The most basic definition that we need to understand is the domino effect itself, which is the successive destabilization of neighboring elements in a regularly spaced array. After understanding the setup of the challenge, we went ahead and began creating a theoretical model in which the only variables we need to define in order to determine the topple time are the number of dominoes, N, and the spacing interval between two dominoes, c. The other variables that are known constants in our set of equations are the following:

- The average width of a single domino, $a = .006m$.
- The average length of a single domino, $b = .043m$.
- The average mass of a single domino, $m = .005kg$.
- The force of gravity, $g = 9.81m/s^2$.
- The mass moment of inertia, $I = \frac{1}{3}m(a^2 + b^2)$.

5 Theory

We have adopted Shaw's theory, which is based primarily on the geometry of interacting dominoes and an energy balance of the system. The first equation we used was

$$\sin(\theta_{N-1} - \theta_N) = \frac{c\cos\theta_N}{b} - \frac{a}{b} \tag{1}$$

where θ_N = the angular position of the final domino. This equation solves for the angular position of the dominoes by using the position of the preceding domino. The coordinate system is described in the figure below. We also looked at the energy of the system, which is

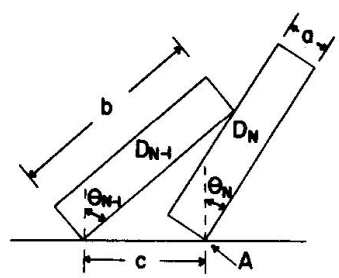

Figure 1: Geometry of two successive falling dominoes, D_N and D_{N+1}

a sum of the total potential energy and total kinetic energy. From this information we can write the energy equation to be solved as part of our theory:

$$E = \frac{1}{2}mg(b\sum_1^N \cos\theta_i + a\sum_1^N \sin\theta_i) + \frac{1}{2}I\sum_1^N \omega_i^2 \tag{2}$$

where θ_i = the angular position of a given domino, ω_i = the angular velocity of that same domino, and I is the mass moment of inertia for a single domino. Using this equation and the initial conditions at time $t = 0$, we were able to obtain a value for the energy of the system.

Since we based most of our theory on the article written by Shaw, we needed to solve for the constants U_i, V_i, and W_i used in Equation (6). To do this, we used the values we obtained from Equation (1) and Equations (3), (4), and (5).

$$\cos\theta_i = U_i\cos\theta_N \tag{3}$$

$$\sin\theta_i = V_i\sin\theta_N \tag{4}$$

$$\omega_i = W_i\omega_N \tag{5}$$

After all of the variables have been solved, we can then solve Equation (2) with the other equations to obtain ω_N as a function of θ_N. This yields Equation (6).

$$\omega_N = \sqrt{\frac{2E - mg(b\cos\theta_N \sum_1^N U_i + a\sin\theta_N \sum_1^N V_i)}{I\sum_1^N W_i^2}} \tag{6}$$

Next, we used an equation to solve for the time it takes one domino to strike the next (t_N), i.e., the time it takes for D_{N-1} to contact D_N.

$$t_N = \int_0^{\theta''} \frac{d\theta_N}{\omega_N} \tag{7}$$

Finally, the cumulative time T_N, defined as the time when the first domino, D_1 strikes D_2 until D_N strikes D_{N+1} was found using the summation formula

$$T_N = \sum_2^N t_i. \tag{8}$$

This equation does not take into account the time for the first domino, D_1, to fall and strike the second domino, D_2. It also does not take into account the time it takes for the final domino, D_N to fully topple. Therefore, these times need to be added to Equation (8) in order to obtain the total topple time of the entire chain of dominoes.

The understanding of this theory is critical to the development of a working simulation. We used Matlab to model our theory in order to obtain concrete results. This code can be found in Appendix A.

First, we applied this theory and the model we developed to a chain of evenly spaced dominoes with the goal of predicting the total topple time. After discovering that Shaw used the same physical set up of dominoes in order to derive his theory, we spent all of our time creating a model that coincides with the theory. When we produced an accurate model based on the theory, the next step was to compare our theoretical data to experimental data collected in the lab. This would verify that our model is accurate and can be further manipulated in order to have correct predictions for different arrangements of dominoes. Figures 2 and 3 are comparisons of the spacing versus topple time for distances of 15cm and 20cm.

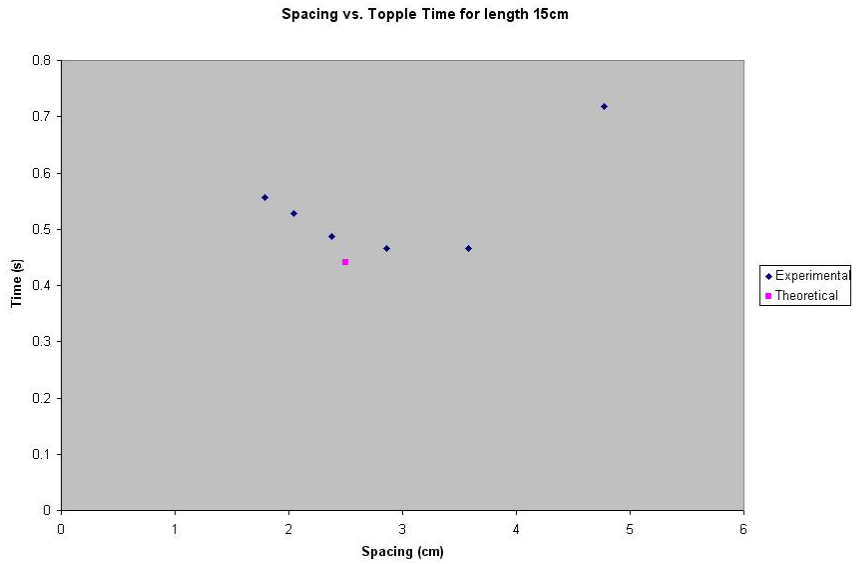

Figure 2: Comparison at 15cm.

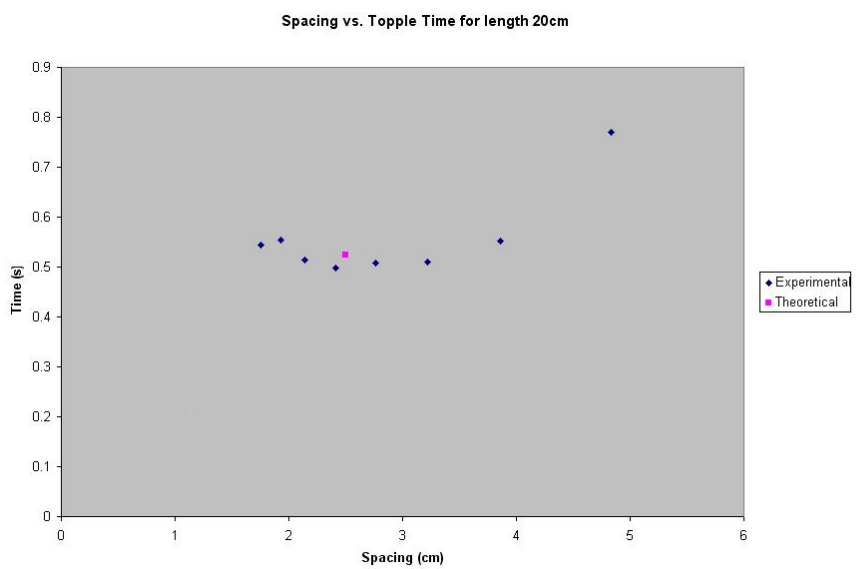

Figure 3: Comparison at 20cm.

Next we altered our code in such a way as to minimize the topple time to the best of our capabilities. Therefore, we will be optimizing a chain of six dominoes since the chain reaches steady-state after $N = 6$, for a spacing of about 2.5cm or less [1]. This can further be explained by examining a chain of dominoes in motion. Due to the overlapping effect that happens when successive dominoes collide, the first domino does not come to a resting position until the sixth domino begins to topple. Physically, this means that there is always a set of six dominoes moving through the chain, and once the first domino is allowed to come to rest, the chain can reach steady state. This setup with six dominoes and a designated spacing will then become a constant in our theory. The spacing of the 6 dominoes was

determined by running experiments of different even spacings of dominoes. Therefore fastest topple will be used in all cases where topple time is to be minimized. Finally, we will then optimize the remaining length in order to achieve the shortest total topple time. After the chain reaches $N = 6$, the spacing is then allowed to change. We will alter this spacing depending on the length, L, that we are given. The value of L after the length the initial six dominoes covers is subtracted and determines how many additional dominoes are needed to cover the specified distance. Our idea is that the best way to minimize the total topple time is to reach the steady state as quickly as possible, and then to choose the fastest equal spacing for the remainder of the distance. The spacing is determined by running the Matlab program we have developed for different spacings, and iterating until the shortest time is found. A comparison of the spacing versus topple time for $N = 6$ can be found in Figure 4 and clearly shows the reason why we chose to set up the first six dominoes with a spacing of 2.5cm.

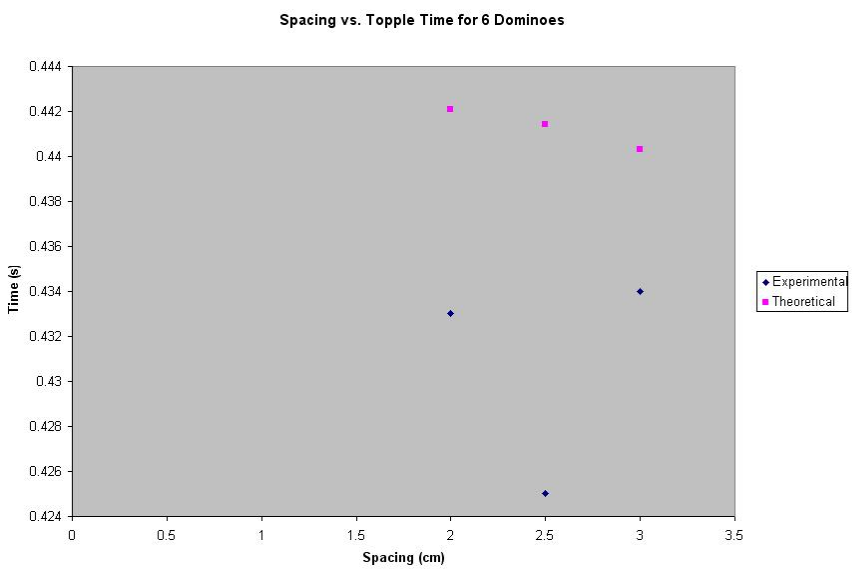

Figure 4: Data for optimization of six dominoes.

For the next competition, we needed to develop a setup of dominoes whose total topple time we can predict with the least amount of error. We will take this arrangement and challenge other teams to try and have a closer prediction than us. During the time we worked, we picked a single domino setup that our model was already successful with and worked to make any possible improvements. We then spent time in the lab, running the same experiment numerous times until we came up with an accurate average and compared this to the results given by our model.

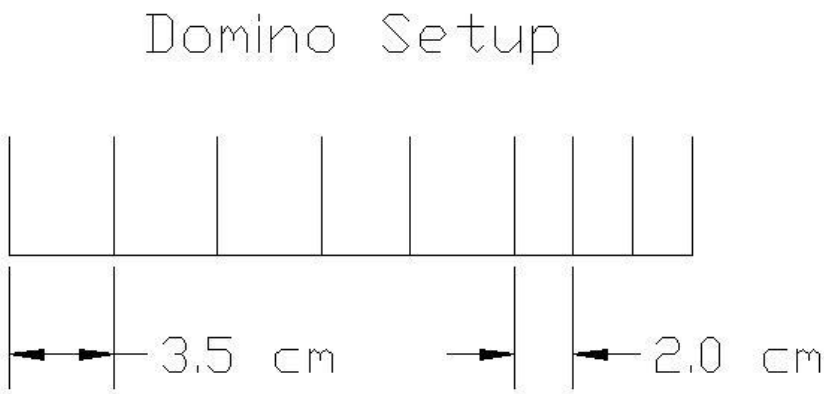

Figure 5: Diagram of domino setup.

The following figure is a comparison of the experimental and theoretical data we had for this particular setup. It shows how close our theoretical prediction came to the experiments we ran in the lab. Since we will be giving this setup to other teams to predict and we will have a lower rating if another team beats us at estimating this setup, it was necessary to have a model that is able to estimate the topple time as precisely as possible. After consistently getting this accurate output from our model, we felt confident enough to use this setup for the third challenge.

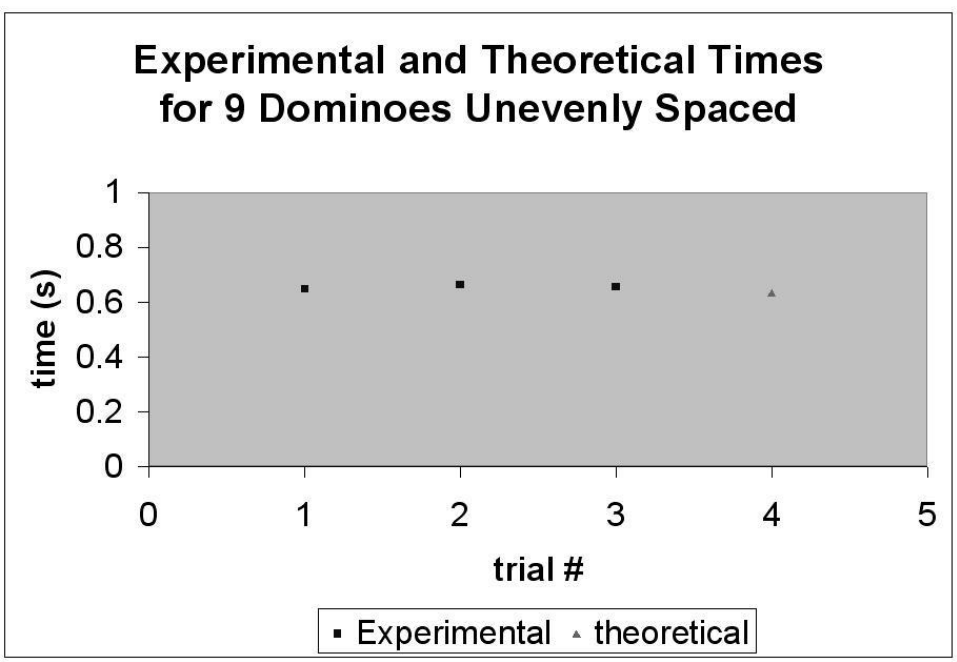

Figure 6: Unevenly spaced domino times.

In the same way, other teams will also be giving us different arrangements of dominoes whose topple time we will need to predict. In order to account for the endless combinations teams can challenge us with, we decided to include an array at the beginning of our model.

This will help us predict the topple time for dominoes that are set up with a different spacing in between each domino.

The final competition will be comprised of 3 individual challenges. The first challenge will be to minimize the topple time given a fixed length and number of dominoes. We will construct our arrangement using the same method as in the second competition where we split up the length and optimized the first section for $N = 6$.

The second and third challenges of this milestone are to predict the topple time of each team's individual setup and Professor Pelesko's, respectively. The difficult part of this challenge is that the arrangement can be non-linear. Based on the constraints of how the dominoes can be arranged, we have been able to determine a method to accurately predict the topple time regardless of whether or not it is a linear arrangement. The constraints are that that the dominoes must be less than 90 degrees between reference corners and each domino must contact the the next successive domino. We will be given the coordinates of each domino in the chain. Using the data provided to us, we will be able to calculate the point at which each domino will contact the next. Figure 7 shows how this effective spacing is calculated. The effective spacings can then be entered into our Matlab code and the output will be the predicted topple time. This method converts a non-linear setup into an effectively linear setup that we can analyze. Our justification for doing this lies in the assumption that each domino always rotates along its front bottom edge and does not rotate in another direction. This assumption allows us to assume that the force always continues in the plane of the topple (parallel to the surface the dominoes are arranged on) , and no energy is lost to other types of domino interactions. We expect that our method will accurately predict the topple time for any arrangement we might be challenged with.

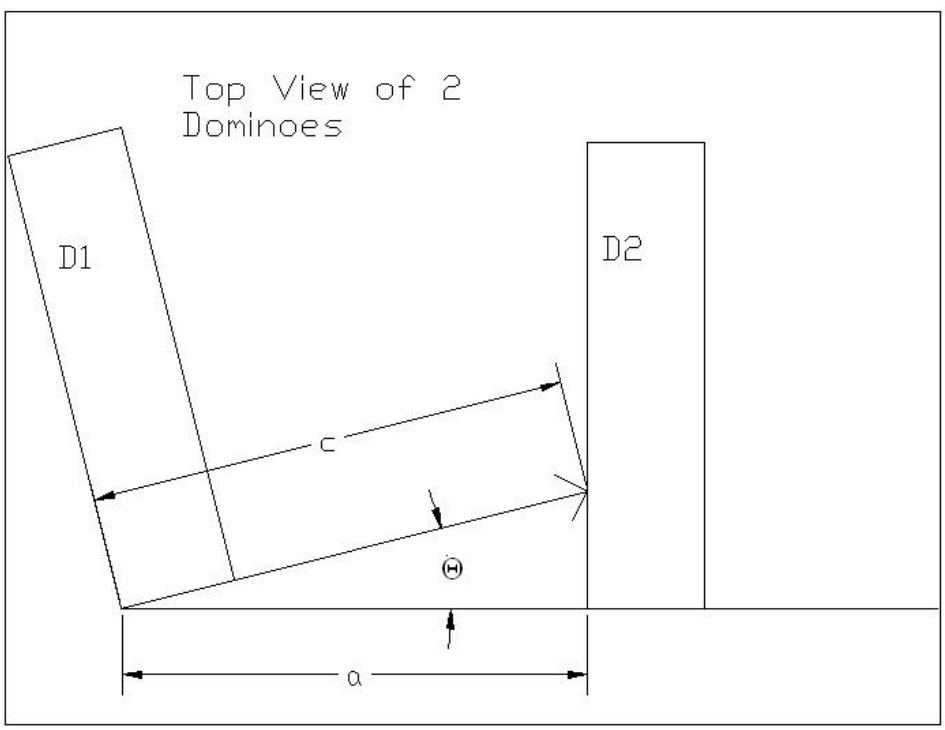

Figure 7: Calculation of effective spacing.

This graph shows the topple time for $N = 7$ dominoes. It is the experimental data collected versus the theoretical prediction that our model outputs. From this graph, one can see that though there will be instances when some experimental data throws off the theoretical prediction, generally our model will be able to handle a case of uneven linear spacing, which was demonstrated in competitions two and three.

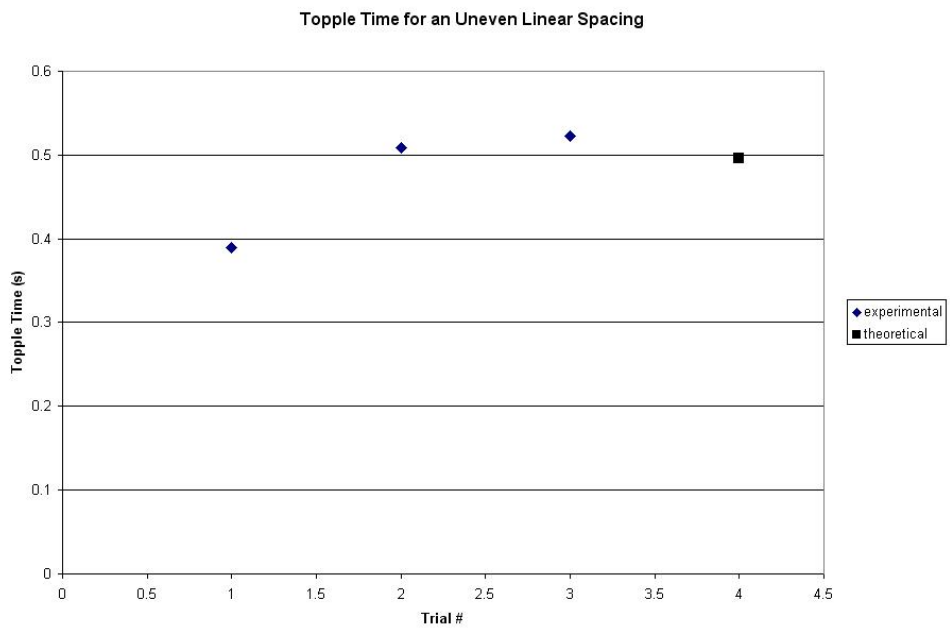

Figure 8: Data for uneven spacing.

Finally, the following is a graph which shows the topple time for a semi-circular chain of dominoes. It includes a visual representation of experimental data versus our theoretical output. By studying this graph, it shows that our way of measuring the distance between the dominoes using the angle that they are rotated at is fairly accurate. Our theoretical prediction is close to two of the experimental data points, though again there is the risk of experimental data skewing the average and in turn making it difficult for our theoretical prediction to coincide with it.

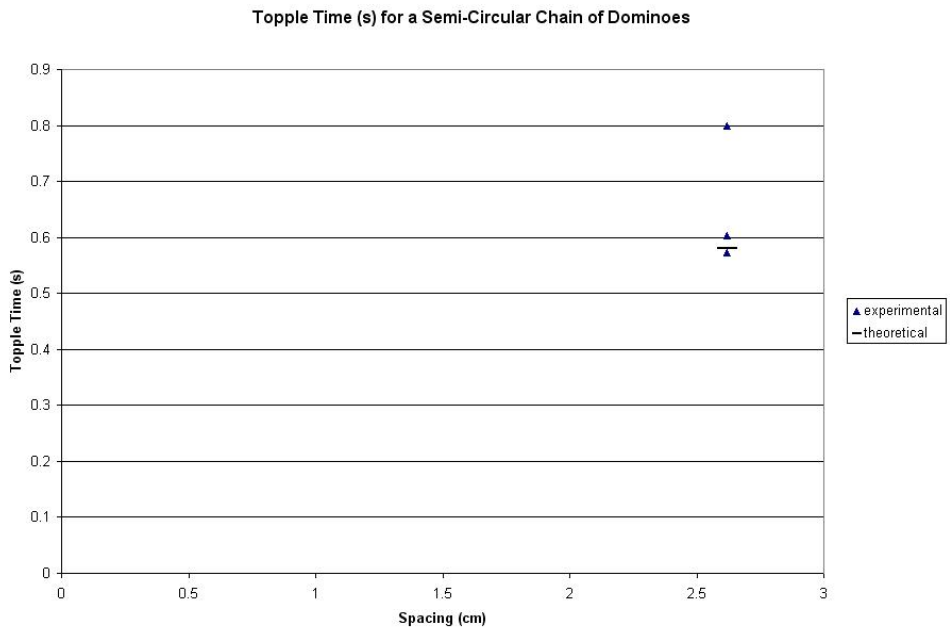

Figure 9: Data for semicircular arrangement.

6 Sample Competition Results

Though this is not a comprehensive listing of all the competition results, it clearly demonstrates the capabilities of our theoretical model. The table below shows that not only were we able to out predict our competition with our own domino setup, but also demonstrates how we can manipulate our model to calculate the topple time for other setups within minutes of receiving the data.

	Our Setup	Their Setup
Our Prediction	0.6336	0.5799
Experimental Outcome	0.6490	0.5717
Other Team's Prediction	0.6745	0.5456

Table 1: Competition Results.

7 Conclusion

Based on the results from all of our experiments and competitions, our prediction method has proven to be a very dependable theoretical model. It has been able to accurately predict the topple time for a chain of dominoes consistently. We are considerably enthusiastic about the abilities of our model due to the fact that it is derived from theoretical information. Also, due to the fact that our predictions are made using a Matlab program, we are able to guarantee the outcomes to be free from human errors during calculation. Due to the need

for optimization of the total topple time, the discovery of the natural propagation discussed earlier became a key aspect in the precision of our model.

8 Annotated Bibliography

8.1 Milestone 1

1. Banks, R. B. (1998). Towing Icebergs, Falling Dominoes, and Other Adventures in Applied Mathematics. Princeton, Princeton University Press.

   In this source, there is much background information on waves and wave motion. This is important material to know since it explains some of the concepts behind falling dominoes. Again, there is much detail about the formulas used in the calculations, including that which is used to determine the time of fall. With the independent variable being the spacing ratio and the dependent variable being a dimensionless parameter, there is a graph comparing the results of the mathematical model with experimental data. Though they are not identical, the graph shows that the two are very similar. This helps convince the reader that the author's mathematical model is sufficient enough for their own use. Also, at the end of the article, the author also recommends two other sources that might be helpful with this topic.

2. Bert, C. W. (1986). "Falling Dominoes." SIAM Review **28**(2): 219-224.

   This article goes into much detail about the derivation behind a set of equations that can be used to solve the variables associated with falling dominoes. It is important to understand the meaning behind all of these equations in order to fully grasp the concept of the problem. Though detailed, the author of the article explains that he did not collect the experimental data used in this report. Since he was not present at the time when the experiment took place, there are a few reasons why one should not rely solely on his article. One reason is due to the fact that the scientists who did complete the experiment did not report on the roughness conditions of the surface on which the experiment was performed, putting one of his assumptions into question. Since he wrote the article under the impression that there are no external friction losses, some of his equations would change if the experiment was performed on a rather smooth surface. This reinforces the fact that information on this topic needs to be gathered from various sources.

3. MacAyeal, D. R., T. A. Scambos, et al. (2003). "Catastrophic ice-shelf break-up by an ice-shelf-fragment-capsize mechanism." Journal of Glaciology **49**(164).

   Assumptions:

   (a) idealized rectangular ice-shelf fragment

   (b) width of ice is infinite and therefore a 2D problem

   (c) the ice slides on the normal surfaces without any shear traction

   (d) force is a function of theta

(e) Center of mass is located at the center of the ice-shelf fragment

(f) Torque of gravity at center of mass is zero

(g) Uniform density

This journal article gave some good insight on how to model the falling dominos. There are some differences between their research and our project. The dominoes are falling through the air and there is no water that must be taken into account. Also, we are not wedging apart the dominoes and breaking them from the ground. There are some governing equations in this source that may become helpful as the project goes on.

4. McGeer, T. and L. H. Palmer (1989). "Wobbling, toppling, and forces of contact." American Journal of Physics **57**(12): 1089-1098.

   This source points out that it is important to realize that the surface we are working with may not be perfect. It may be necessary to take into account that our ground is not perfectly level. There may also be wobbling of the domino that would need to be accounted for. This article also looks at the slippage before the topple. We are using sandpaper as the surface, so this effect will be minimized and possibly ignored.

5. Perkins, S. (2002). "Toppling icebergs sped breakup of Larsen B ice shelf." Science News Online **162**(25/26). Sachtjen, M. L., B. A.

   This was a very short article that basically pointed out that the domino effect can be very powerful. One change can have very widespread effects.

6. Sachtjen, M. L., B. A. Carreras, et al. (2000). "Disturbances in power transmission systems." Physical Review E **61**(2): 4877-4882.

   This source deals with power transmission systems, but may be applied to the domino problem in this class. The article says that there are three types of elements in the system: power generators, power consumption centers, and transmission lines. The domino project may be able to be divided similarly. Also, some information about the probability of disturbances may be gleamed from this source.

7. Shaw, D. E. (1978). "Mechanics of a chain of dominoes." American Journal of Physics **46**(6): 640-642.

   In the "Mechanics of a chain of dominoes," Shaw discusses an experiment, much like competition 1, involving a linear chain of dominoes. Shaw explains that sandpaper is used to ensure that "the point of contact between falling dominoes and the surface does not move." "Mechanics" discusses a theoretical model of a chain of dominoes for computing Tn, which is the time interval between the first domino striking the next, and so on until the last domino falls. This value can be computed by integrating the angular velocity of the last domino. "Mechanics" will be incredibly useful for our work in the lab during competition 1. It will be useful in computing the time it take for all the dominoes to topple, using the formula

$$\sin(\theta_{n-1} - \theta_n) = \frac{(c)\cos(\theta_n)}{b} - \frac{a}{b} \tag{9}$$

for measuring a relation between angles of dominoes, where θ represents the angular positions of the dominoes and a, b, and c are the dimensions of the dominoes.

8. Stronge, W. J. and D. Shu (1988). "The domino effect: successive destabilization by cooperative neighbors." Proceedings of the Royal Society **418**: 155-163.

   Stronge and Shu define the "domino effect" as the "successive destabilization of neighboring elements in a regularly spaced array." It is also stated that the natural speed of toppling dominoes increases as the number of dominoes increase. "Successive" explains that the natural toppling speed is larger for a "cooperative group" than for a single collision "unless the loss of energy at each collision is small and the spacing between the elements is large." Certain features of toppling can be considered if we assume the following,

   (a) There is an indefinitely large number of toppling dominoes behind the collision wave front;

   (b) Each domino behind the wave front leans forward against a neighbor;

   (c) During toppling, friction at the sliding contact between dominoes is negligible;

   (d) the coefficient of restitution is negligible; hence, after a collision, colliding blocks remain in sliding contact.

   From these assumptions we can take the relative angular speed to be:

$$\frac{\theta_{i+1}}{\theta_{\ i}} = 1 - \frac{\lambda + h}{L} \frac{\sin(\theta_i)}{\cos(\theta_{i+1} - \theta_i)} \tag{10}$$

   Where θ_i is the rotation of domino i and λ is the distance between dominoes and h and L are the dimensions of the domino. "Successive" also explains that studying the "domino effect" can be useful to describe events such as "sequential collapse of periodic frame structures," like the Larsen A ice shelf we discussed in class.

9. Stronge, W. J. (1987). "The domino effect: a wave of destabilizing collisions in a periodic array." Proceedings of the Royal Society **409**: 199-208.

   (a) The collision wave front moves through the array at a speed asymptotically approaching an intrinsic speed that depends on the spacing and friction.

   (b) The essential features of the "domino effect" are examined in collisions if the blocks are initially at rest and have a small positive stability.

   (c) During toppling of closely spaced dominoes, the initial momentum is not regained, so the total number of toppled dominoes depends on the initial momentum passed through the system.

8.2 Milestone 2

1. Shaw, D. E. (1978). "Mechanics of a chain of dominoes." American Journal of Physics **46**(6): 640-642.

   For this milestone, we spent most of our time reviewing this source. It goes into much detail describing the theory behind solving the topple time for a chain of equally spaced dominoes. Since this is the basic setup of the first challenge, we decided to dedicate the majority of our time to understanding and trying to apply Shaws theory. While studying Shaws theory, it becomes evident that an analytical solution will be difficult to reach and that a numerical solution must be found. In order to do this, we have written a program in Matlab which will require an input for N, the number of dominoes in the chain being tested, and c, the spacing distance between the dominoes. This will allow us to solve for the topple time of the entire chain during the challenge.

2. Mathews, John H. and Kurtis D. Fink. Numerical Methods Using Matlab. 4th ed. New Jersey: Pearson Prentice Hall, 2004.

   This piece of literature is the textbook used in the University of Delawares Math 353 class. It is a very beneficial reference due to the fact that it includes various techniques used in writing Matlab programs. Since we will be using a program to compute the topple time of the chain of dominoes during the challenge, it is important to create a working program that can handle the extensive formulas included in Shaws article.

9 Appendix A